

The Riemann Zeta Function: Zeros, the Critical Line, and Computational Exploration

Computational visualization, prime structure, and experimental
mathematics

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Roadmap

- 1 Introduction
- 2 Computation
- 3 Conclusion

From Euler to Riemann

Euler's resolution of the Basel problem revealed that the zeta function already encodes remarkable arithmetic structure at special values such as

$$\zeta(2) = \frac{\pi^2}{6}.$$

Riemann's later insight extended this perspective dramatically: after analytic continuation, the non-trivial zeros of the zeta function govern the fine-scale distribution of prime numbers.

The Riemann Zeta Function

- The Riemann zeta function occupies a central place in analytic number theory.
- For $\Re(s) > 1$, it is defined by the convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

- Euler showed that this function also admits the product expansion

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1},$$

which connects analysis directly to the prime numbers.

Central Thesis

Big idea

The zeros of $\zeta(s)$ are not merely analytic curiosities. They are structural features that govern the oscillatory behavior of prime counting.

This viewpoint culminates in the Riemann Hypothesis, which asserts that every non-trivial zero lies on the critical line

$$\Re(s) = \frac{1}{2}.$$

Prime Number Distribution

This broader perspective also situates related phenomena in prime number theory.

- **Mersenne primes:** numbers of the form $2^p - 1$, where p must be prime for $2^p - 1$ to have any chance of being prime.
- **Twin primes:** pairs of primes differing by 2, such as $(29, 31)$.
- The Twin Prime Conjecture asserts that infinitely many such pairs exist.

Where the Zeros Live

The non-trivial zeros of the Riemann zeta function lie in the critical strip

$$0 < \Re(s) < 1.$$

Extensive numerical evidence suggests that these zeros lie on the critical line

$$\Re(s) = \frac{1}{2},$$

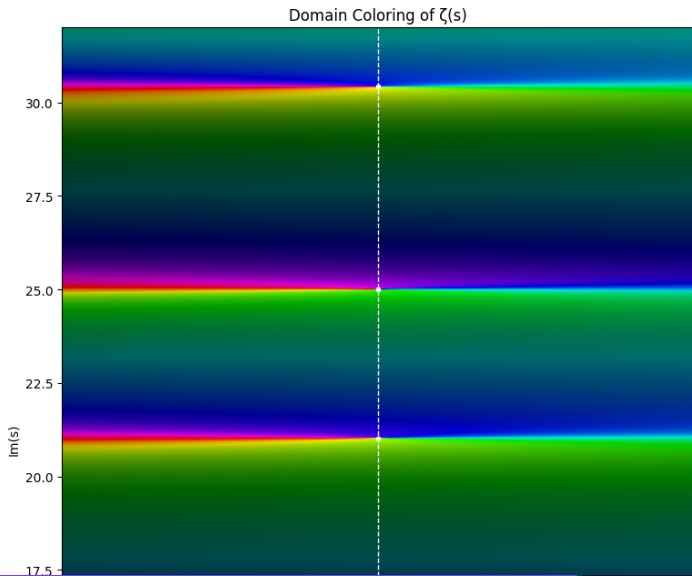
as asserted by the Riemann Hypothesis.

Visualization techniques provide useful geometric intuition for the analytic structure of $\zeta(s)$ in the complex plane.

Visualization Modes

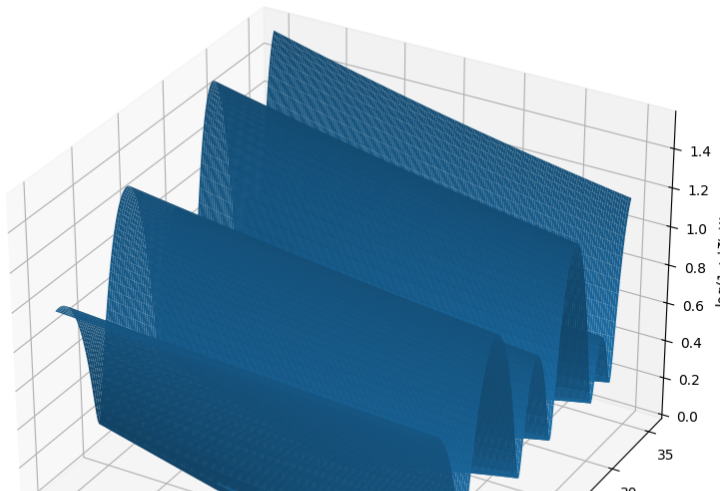
- **Phase Plots:** hue tracks $\arg(\zeta(s))$ while brightness tracks $|\zeta(s)|$; zeros appear as phase singularities.
- **Contour Plots:** the curves $\Re(\zeta(s)) = 0$ and $\Im(\zeta(s)) = 0$ intersect exactly at zeros.
- **Three-Dimensional Surfaces:** the graph of $|\zeta(s)|$ creates a landscape whose minima align with non-trivial zeros.

Phase Graph



Surface Plot

3D Surface Plot of $\log(1 + |\zeta(s)|)$



Takeaway

These visual representations reinforce the interpretation of the zeros as structural features governing the behavior of $\zeta(s)$, and by extension, the distribution of prime numbers through the explicit formula.

Computational Consequences

- **Computational Evidence:** extensive computations have verified enormous numbers of non-trivial zeros on the critical line.
- **Prime Counting:** the zeros appear explicitly in formulas such as

$$\pi(x) = \text{Li}(x) - \sum_{\rho} \text{Li}(x^{\rho}) + \text{lower-order terms}.$$

- **Error Bounds:** RH implies the near-optimal error term

$$\pi(x) = \text{Li}(x) + O\left(x^{1/2} \log x\right).$$

The Big Formula

A central bridge between the zeros of $\zeta(s)$ and the distribution of primes is the explicit formula

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}),$$

where $\psi(x)$ is the Chebyshev function and the sum runs over non-trivial zeros ρ of $\zeta(s)$.

This formula shows that the zeros contribute oscillatory corrections to the main term x .

- **Numerical contour analysis:** visualize level sets of $\Re(\zeta(s))$ and $\Im(\zeta(s))$, approximate zeros, and study local spacing.
- **Automated theorem search:** machine-assisted systems can explore large proof spaces and intermediate lemmas.
- **Large-scale verification:** distributed computation continues to test zero locations to increasing heights.

Classical vs Modern Ciphers

The contrast between classical and modern cryptography highlights why prime numbers matter outside pure mathematics.

Classical example

A Caesar cipher shifts each letter by a fixed number of positions. A slightly more complex variant uses a keyword substitution alphabet such as

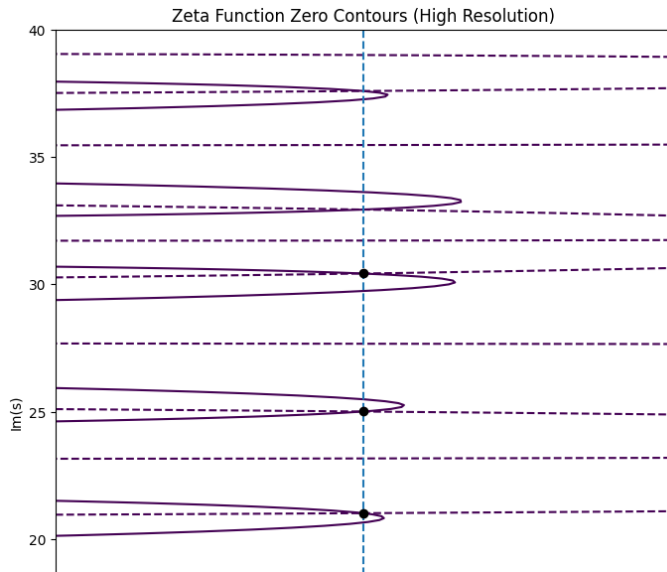
KEYWORDABCFGHIJLMNPQSTUVXZ.

These systems are weak because they leak repeated letters, frequency patterns, and key-space structure.

Cipher Comparison

System	Key Space Size	Attack Method	Time to Break
Caesar Cipher	25 keys	Brute force	Instant (< 1 ms)
Keyword Substitution	Large theoretical space	Frequency analysis	Seconds–minutes
DES (56-bit)	2^{56}	Exhaustive search	Hours–days
RSA-512	Large composite modulus	Integer factorization	Hours–months
RSA-2048	Very large composite modulus	Integer factorization	Infeasible classically

Contour Plot



Conclusion

- The Riemann Hypothesis links complex analysis with the fine structure of prime distribution.
- Euler opened one door into the zeta function through special values.
- Riemann opened another through the zeros, which govern oscillatory structure in the primes.

Why the Visuals Matter

The contour plot makes the geometry concrete:

- intersections of $\Re(\zeta(s)) = 0$ and $\Im(\zeta(s)) = 0$ reveal non-trivial zeros;
- the explicit formula ties those zeros directly to prime irregularity;
- computation and visualization create a practical way to explore deep analytic structure.

Future Work

Future work will extend the computational side of this project in four main directions:

- high-precision numerical approximation of $\zeta(s)$ along the critical line;
- richer visualization, including domain coloring and surface plots;
- experiments with truncated explicit formulas for $\psi(x)$ and $\pi(x)$;
- exploratory AI-based analysis of zero spacing, clustering, and prime-related data.